

A virtual centrifugal force based navigation algorithm for explorative robotic tasks in unknown environments

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Abstract

In many robotic tasks, there is no a priori knowledge of the environment. This makes it necessary for robots to explore the environment. Navigation algorithms for robots to map the environment completely in a short time play a very important role in the robotic task completion. A navigation algorithm based on virtual centrifugal force is proposed to complete the robotic exploration of the unknown environment using rang sensors in this paper. Collisions between a robot and an obstacle or between robots can be avoided with the application of the proposed navigation rules. The kinematics and dynamics equations of robots adopting the algorithm are also given. The simulation experiments demonstrate the operation of the algorithm. Several simulation experiments of various representative robotic tasks are carried out, based on the explorative navigation algorithm, which successfully validate the virtual centrifugal force based navigation algorithm.

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1. Introduction

In many robotic tasks, environments are partially known, which requires robots to acquire a spatial model of the environments. Maps are commonly used for the robot navigation. To acquire a map, the robots have to possess sensors that enable them to perceive out-

side world. The sensors commonly used for this task include cameras, range finders, radar, tactile sensors, compasses and GPS [1]. Most sensors are subject to range limitations so that a robot has to be navigated through its environment when building an environment map. Generally, navigation includes path planning in the already known environment and exploration in the unknown environment. In the former case, the problem of the path planning becomes a geometric design with the environment model as an input to the system. As for the navigation under the latter condition, the robots are required to acquire on-line real-time information using

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specific sensors, to plan a collision-avoidance path, and to carry out other operations [2]. Especially for explorative application tasks, the robots have to build reliable maps in advance for their task environment in virtue of appropriate sensors and navigation algorithms. With appropriate sensors, the robot can obtain local environment information that can be accumulated and extended by the use of a good navigation algorithm till the whole environment is mapped.

In the case where the environment is unknown, on-line real-time exploration algorithms, which use local sensing information, are needed [3,4]. During the navigation, the robots are required to have the following capabilities: (a) detect the main features of the environment and describing them in an effective form; (b) avoid collisions between robots and obstacles; (c) avoid repeated detection as much as possible. According to the sensor in use, the previous navigation algorithms can be classified into two types: vision based ones and non-vision based ones. Visual sensor based navigation algorithms include: *spurs* based algorithms by Deng et al. [5] and Sutherland [6], a paradigm based on *navigation course* ξ by Rao [7], VisBug-21 and VisBug-22 by Lumelsky and Skewis [8], *sightseer strategy* and *seed spreader strategy* by Lumelsky et al. [9]. Navigation algorithms not using vision can be classified into metric algorithms and non-metric ones. The metric algorithms include: *Bug-1* and *Bug-2* algorithm by Lumelsky and Stepanov [10], *Alg1* and *Alg2* by Sankaranarayanan and Vidyasagar [11], and so on. All these metric algorithms make use of information such as position and distance. *Curve1* algorithm [12] is an example of the non-metric algorithms. A good overview on non-heuristic path planning algorithms is given in [2]. The main purpose of most algorithms in [2] is to enable a robot to arrive at some preset targets from a start position without collisions. Generally, the lesser distance it covers, the better the strategy is. For explorative and mapping tasks, however, these algorithms are inapplicable due to the requests that are different from the ones in path planning tasks.

When the robot navigates in an unknown environment, it is often required to build a reliable map of the environment at the same time. Research on map-building can be classified into two domains according to the sensor type. One is based on range sensors and the other is based on vision sensors. The latter usu-

ally combines the range sensors and vision techniques together.

On the environment exploration and mapping, a robot guide named Minerva at Carnegie Mellon University [13] has mapped a museum by virtue of Monte Carlo localizer. Thrun et al. [13] have done much research work on this topic which focuses on algorithms of eliminating localization errors, such as Expectation Maximization, Bayes filters, Markov methods, Kalman filters, extended Kaman filter, and so on. Pagac et al. [14] and Cai et al. [15] have carried out experiments separately on single and multiple robotic mapping of simple environments using range sensors. Both of them employed the evidential reasoning theory to eliminate the uncertainty of the information collected by the range sensors. Hans et al. [16] has introduced a metric algorithm for stochastic mapping, which generalizes the extended Kalman filter to incorporate vehicle localization and environment mapping. Araújo and de Almeida [17] have proposed an on-line predictive trajectory method based on a fuzzy ART neural architecture to update environment mapping. Duckett and Nehmzow [18] have presented a map-based exploration system using an artificial neural network to detect new areas of an unexplored territory. Yamauchi [19] has proposed a frontier-cell method for exploration. Shanahan and Witkowski [20] has presented a programmable logic based agent control system that interleaves planning, plan execution and perception with event calculus. His research emphasis is on how to describe the relationship between the agent and its environment. In the research of Moorehead et al. [21], a search algorithm has been used to collect information. Thrun and Bücken [22], Rekleitis et al. [23] and some other researchers have integrated the grid-based mapping into the topological maps extracted from grid ones for the robot navigation.

On exhaustive exploration, Spires and Goldsmith [24] adopted the Hilbert space-filling curve (Fig. 1) in their geographic search by mobile robots. However, the method requires that the region subdivided into 2^{2D} subregions, where the integer D is the curve order and the robots' initial positions can be placed only in the corner subregions. The routes of robots are predefined according to U-shaped Hilbert curve. So the navigation is not strictly real-time.

On the basis of the previous work in robotic mapping and navigation, an on-line navigation algorithm using

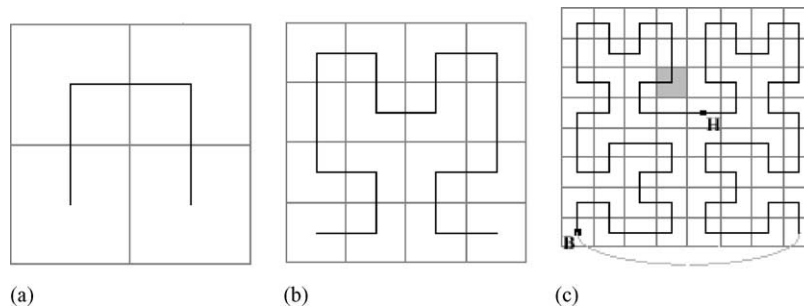


Fig. 1. Exhaustive geographic search along Hilbert space-filling curves: (a) first-order Hilbert curve; (b) second-order Hilbert curve; (c) third-order Hilbert curve.

virtual centrifugal force is proposed for environment exploration tasks. With this navigation algorithm, the robot equipped with sonar range sensors can acquire environment information in real-time and build a metric map integrally. Since the robot has no prior experience of knowledge learning, the method is non-heuristic.

The paper is organized as follows. After a survey of related work, the algorithm and some suppositions are detailed in Section 2. Several experiments on different tasks are simulated to demonstrate the operation of the virtual centrifugal force based navigation algorithm. Finally, some conclusions and discussions are given.

2. Virtual centrifugal force based navigation algorithm

2.1. Preliminaries

Before the introduction of the proposed algorithm, the following assumptions are made:

- (1) The task environments are modelled and represented in a grid mode. Each cell (i, j) of the environment has a value $env[i][j] \in \{0, 1\}$, where 0 means that the cell (i, j) is free, while 1 represents that cell (i, j) is occupied with an obstacle.
- (2) Sonar range sensors are idealized, so it will not be disturbed by noise or errors caused by physical structure. Although this assumption may be slightly different from practical applications, it will simplify some unimportant questions in research. In real experiments, effective emendation measures are necessary to eliminate the noise or errors of the physical sensor.

- (3) As for the inherent measurement uncertainty in the range sensor, Dempster–Shafer theory based evidential reasoning with uncertainty is employed to eliminate it. With Dempster–Shafer theory based information fusion, the acquired map can be updated incrementally.
- (4) The robot occupies a cell and knows its initial positions. It can also calculate the next position in terms of the current position and motion direction. During the simulation, odometry localization errors caused by the physical robot and metrical instrument are not taken into account.

2.2. Description of the algorithm

The idea of the virtual centrifugal algorithm is illuminated by the particle moving in motion of Archimedes spiral. When a particle winds around a fixed center at a continuously increasing or decreasing distance from that point, the dynamic swirl surroundings will extend a swirl source center to a swirl area larger and larger. Following the spiral principle of swirl expanse mode, a new robot navigation algorithm is developed for the robotic mapping task. In explorative tasks, the initial position of a robot can be set near to the fringe of the environment or at any other area of the environment. According to the principle of spiral motion, the robot begins its detection from its initial position like a swirl source, moves gradually surrounding the detected area named “swirl center”, and extends the detected “swirl area” bit-by-bit.

The sonar range sensors are installed on the robot by proportional spacing, which results in a total sensing range of 360° and forms a circular sensing area with the robot at its center. The robot extends its exploration

Table 1
Rules for selecting moving direction

$sen\ inf(C(r, step, \pi/2))$	0	0	0	0	0	0	1	1	1	1	1
$rec(C(r, stepd, \pi/2))$	0	1	1	1	1	1	—	—	—	—	—
$sen\ inf(C(r, step, 0))$	—	0	0	0	1	1	1	1	0	0	0
$rec(C(r, stepd, 0))$	—	0	1	1	—	—	—	—	0	1	1
$sen\ inf(C(r, step, \pi/2))$	—	—	0	1	0	1	0	1	—	0	1
$F_{virtual}$	F_{cen}	0	$-F_{cen}$	$-\sqrt{2}F_{cen}$	$-F_{cen}$	$-\sqrt{2}F_{cen}$	$-F_{cen}$	$-\sqrt{2}F_{cen}$	0	$-F_{cen}$	$-\sqrt{2}F_{cen}$
Resulting direction	$\alpha + \pi/2$	α	$\alpha - \pi/2$	$\alpha - \pi$	$\alpha - \pi/2$	$\alpha - \pi$	$\alpha - \pi/2$	$\alpha - \pi$	α	$\alpha - \pi/2$	$\alpha - \pi$

to the periphery of the explored area with the initial position as its reference. Ultimately the resulting motion demonstrates a tendency of some spiral types. What causes a particle to move in a spiral fashion is an effect attributed to the centrifugal force. Similarly, an imaginary virtual force $F_{virtual}$ is introduced where there is a change of the robot motion direction. This virtual force is always perpendicular to the current moving direction of the robot. A positive virtual force means that it is towards the detected area, and a negative one that it is deviation from the detected area while toward the undetected periphery.

The robot decides the direction of its next step in the term of Table 1. The five types of environment information are checked in the order of the line ranks in Table 1. At each step, there are four possible directions for the robot to select from. According to the proposed spiral trochoid, the priority of the four directions can be set from high to low as follows: turning right at 90° , keeping the direction unchanged, turning left at 90° , turning left at 180° .

In Table 1, $C(r, d, \theta)$ represents the computed coordinates that is at a distance of d from the robot whose current position is at $(r.x, r.y)$. $C(r, d, \theta) = (r.x + d \times \cos(\alpha + \theta), r.y + d \times \sin(\alpha + \theta))$, where α denotes the direction of the robot at its current step i.e. the angle measuring from x -axis to robot moving direction; θ the departure angle from the moving direction, which is positive at clockwise and negative at counter-clockwise. Variable $sen\ inf(i, j)$ records the information value of the detected cell (i, j) , and indicates the occupancy state of the cell; its value will be 0 if the cell is free and 1 if occupied. Variable $rec(i, j)$ records whether the cell (i, j) has been detected; 1 for detected and 0 for undetected. Variable $step$ represents the step length for robot to move from the current position to its candidate position. Variable $stepd$ represents the distance of the center of the neighbor area from the robot, which is ad-

jacent to the current sensing area. Suppose the sensing radius of the robot is SR , then there exists the following relationship: $step < SR < stepd$. Symbol “—” in Table 1 means that information has no influence on the robot decision-making and does not need to be checked. F_{cen} is the centrifugal force which allows the robot to turn right at 90° . The rules listed in Table 1 are for the right spiral; the sign of addition and subtraction in the resulting motion direction will be opposite in the case of the left spiral.

Fig. 2(a) is a graphic illustration of the VCF navigation rules i.e. robot chooses its direction by judging the corresponding information of the ordinal numeration cells. Fig. 2(b) is an example to illustrate the robot tracks adopting the above rules. In the example, $step = 1$, $SR = 2$, and $stepd = 3$.

When the robot finds itself inside an area which is looped with explored cells or with obstacles and explored cells, it will walk out of the area and begin another exploration spiral from a new initial position.

For each step, the robot generates its moving direction of next step in terms of the rules listed above, and the effect appears as the detected area expands to the periphery in a spiral manner. From the above decision-making procedure it can be concluded that this navigation algorithm has satisfied the basic requirements of the robot exploration, i.e., (1) possible collisions with obstacles can be avoided due to information $sen\ inf(i, j)$, from which the robot knows the environment information around itself; (2) repetitive exploration of the same area can be avoided to a large extent due to information $rec(i, j)$ since $rec(i, j)$ records whether an area has been explored; i.e. if an area is recorded as “explored”, the robot will not detect it again; (3) exploration integrality can be fulfilled due to information $rec(i, j)$, by which the robot knows whether all the areas have been recorded as “explored”. Since the robot explores outwards along the fringe belt of the detected

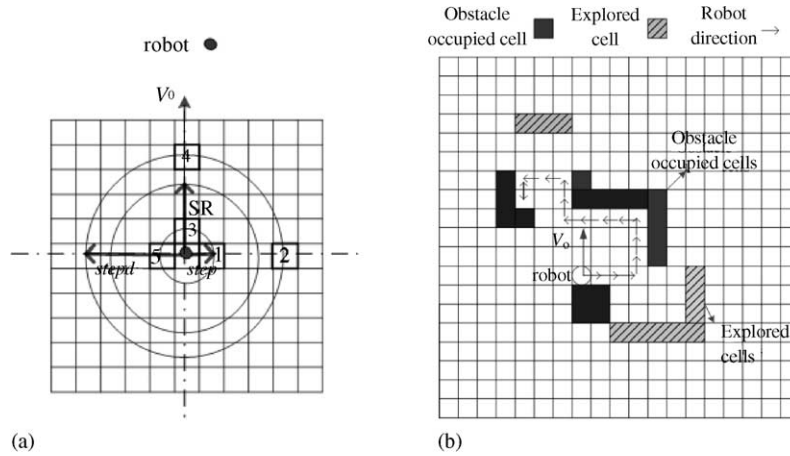


Fig. 2. Illustration of the VCF navigation rules. (a) Graphic illustration of the rules. (b) Example of the robot direction choice adopting the VCF rules.

interior area, the inside accessible area surrounded by the out spiral ring, which is generated as the robot trochoid, is completely detected. Consequently, when the robot terminates its exploration, no unexplored accidental sections are left inside the available area. As the detected areas are continuous, this guarantees the integrality of the exploration. The wall-following strategy [25] is a special case of the proposed algorithm when the robot approaches an obstacle.

When an accident occurs during the exploration, the robot is forced to leave the exploration boundary, and an area may be left undetected while it is abounded by neighboring detected area. However, that does not affect the exploration strategy since the robot may explore in many spirals with different start positions till the whole environment is detected entirely.

2.3. Kinematics equations of robots adopting the VCF algorithm

For robots operating on a planar surface, the pose of the robot is usually modelled by three variables: the Cartesian coordinates in the plane, and the heading direction. According to the rules described in Section 2.2, the robot pose at each step can be expressed as

$$P(k) = (x(k), y(k), \beta(k)), \quad k = 0, 1, 2, \dots \quad (1)$$

where $P(k)$ is the robot pose at the k th step; $x(k)$ and $y(k)$ the abscissa and ordinate values at the k th step, respec-

tively; $\beta(k)$ the heading direction or the angle measured from the ordinate to the motion direction of the robot at the k th step, whose sign is positive at counterclockwise and negative at clockwise, and whose value is an integral of $\pi/2$. With the following recursive formulations the current Cartesian coordinates and heading angle of the robot can be determined from its prior pose at the $k-1$ -th step:

$$\begin{cases} x(k) = x(k-1) + (-1)^{\text{int eg}(n/2)} \text{step} \cos(\beta(k-1)) \\ y(k) = y(k-1) + (-1)^{\text{int eg}(n/2)} \text{step} \sin(\beta(k-1)) \\ \beta(k) = \beta(k-1) - (n-1) \frac{\pi}{2} \end{cases}, \quad k = 1, 2, 3, \dots \quad (2)$$

The initial pose of the robot is given by

$$\begin{cases} x(0) = x_0 \\ y(0) = y_0 \\ \beta(0) = \beta_0 \end{cases} \quad (3)$$

In Eq. (2), $\text{int eg}()$ is a function to get the integer part of its expression. The value of n in the equation is taken as in Table 2.

According to the VCF algorithm at each step, the priority orders of the direction candidates from high to low are: right, forward, left, and backward. Sign “+” represents that the corresponding direction is selected; sign “−” represents that the corresponding direction

Table 2
n Values

Direction candidate	Right	Forward	Left	Backward
<i>n</i>				
0	+			
1	–	+		
2	–	–	+	
3	–	–	–	+

is unable to be selected; while the blank means the corresponding direction does not need to be considered in the case that some direction has been selected. It can be seen that the value of *n* in each row equals to the number of negative sign “–” for the direction candidates.

With the above kinematics equations, initial position and pose, the position and pose of the robot at each step can be computed and its coordinate values are represented as integers.

2.4. Dynamics equations of robots

Following the dynamics theory, the dynamics equation of a robot at the *k*th step can be obtained i.e. the proposed virtual centrifugal force $\vec{F}_{\text{virtual}}$ can be expressed as

$$\begin{aligned}\vec{F}_{\text{virtual}} &= m \cdot \vec{a}_k = m \cdot \left(\frac{\vec{V}_k - \vec{V}_{k-1}}{k - (k-1)} \right) \\ &= m \cdot (\vec{V}_k - \vec{V}_{k-1}), \quad k = 1, 2, 3, \dots, \quad (4)\end{aligned}$$

where *m* is the mass of the robot; $\vec{a}_k$ the acceleration vector at the *k*th step; and $\vec{V}_k$ the velocity vector at the *k*th step. The speed magnitude of the robot is a constant, denoted as V_c . What changes in the velocity vector is the heading direction i.e. the moving direction angle $\angle\beta(k)$. Since a vector can be expressed by its magnitude and heading direction, the velocity vector of the robot is expressed as

$$\vec{V}_k = V_c \angle\beta(k), \quad k = 0, 1, 2, \dots \quad (5)$$

So, Eq. (4) can be rewritten as

$$\begin{aligned}\vec{V}_k &= m \cdot V_c \cdot (\angle\beta(k) - \angle\beta(k-1)), \\ k &= 1, 2, 3, \dots \quad (6)\end{aligned}$$

3. Simulation experiments

Environments for simulation experiments are modelled and depicted in occupancy grids. The total scale is 300×300 . The value of each grid is 0 or 1. 0 represents the free grid and 1 represents the occupied grid with obstacle. To sense the environments, the sonar range sensors are adopted with a radius $SR = 8$ and sensing angle of 15° . Totally, 24 sonar sensors are installed on a robot, and evidential reasoning with uncertainty is used to fuse and update the acquired local information. During all the simulation experiments, robots move in a right spiral mode

3.1. Simulation experiment 1: comparison experiments

To testify the validity of the proposed navigation algorithm, comparison experiments on the virtual centrifugal force (VCF) based algorithm and on a random search algorithm with changeable step length are carried out. A primary disadvantage of the random search algorithm is that terminative conditions are difficult to control. Because the detected regions are discontinuous, it is hard for the robot to cover the entire environment. The environment for the robot to explore is shown in Fig. 3(a), where the black blocks represent the obstacles and the blank means the free space. This environment is also used for other experiments later.

Fig. 3(b) shows the resulting map after the robot explores more than 50,000 steps. Clearly, it is difficult for the robot to explore the environment completely without an appropriate navigation strategy. Only by prolonging the detecting time and increasing the step number is it possible for the robot to detect the basic features of the environment. With the VCF based navigation algorithm presented in this paper, however, almost all the basic environment features can be mapped, as shown in Fig. 3(c). Fig. 3(c) shows the result map generated after a complete spiral exploration. To demonstrate the “swirl” effect of the algorithm more clearly, we put the robot near the middle section of the environment as its initial position, $r_0 = (150, 142)$. The record of the step number shows that the robot walks less than 7000 when it finishes the exploration as shown in Fig. 3(c). Fig. 3(d) depicts the path trace of the robot. The grey areas in Fig. 3(b) and (c) represent the undetected spaces, including the reachable and unreachable

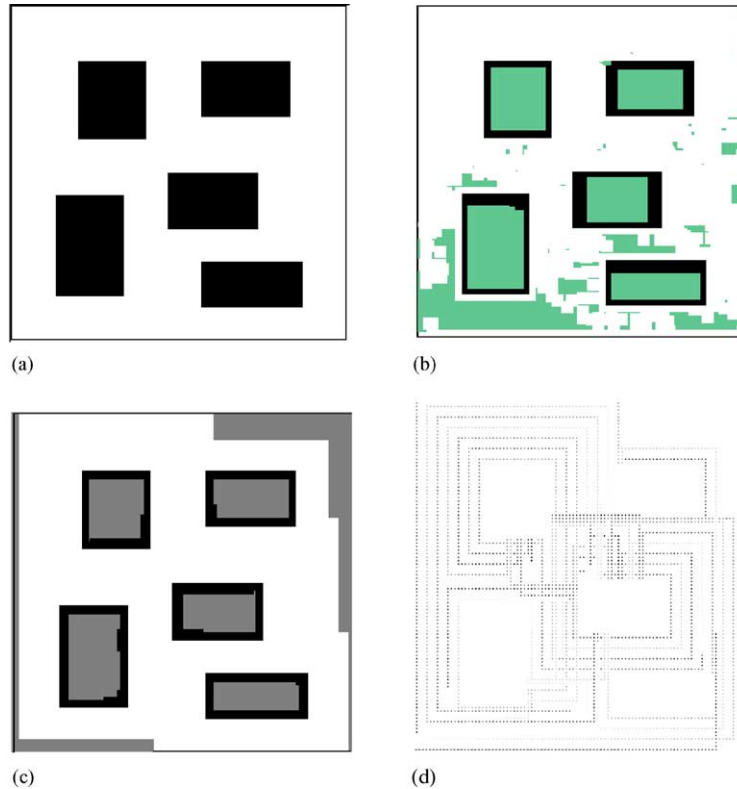


Fig. 3. Comparison between exploration with the VCF navigation algorithm and exploration with the random search scheme. (a) Environment to be explored. (b) Map generated by a robot with random search scheme. (c) Map generated by a robot after completing a whole spiral. (d) Tracks of the robot after completing a whole spiral.

grids. In Fig. 3(b), there exist many discontinuous grey blocks in the available space. While in Fig. 3(c), there leaves no available space undetected inside the outmost fringe formed by the robot trace. It can be concluded clearly that the exploration efficiency and reliability can be improved largely by the use of the VCF based navigation algorithm.

3.2. Simulation experiment 2: exploration with random initial positions

In case that the robot starts the exploration near the environment boundary using the right spiral typed VCF based navigation algorithm, it will expand its exploration approaching the central section from the outer orbicular layer counterclockwise. The resulting trace will be inverse, compared to the case where the robot starts from the central section, that is, the ex-

ploration mode appears from outer loop layer to inner layer. Fig. 3(a) is still taken as the experiment environment. Fig. 4(a) shows the resulting exploration map and the robot track when the robot has an initial position at $r_0 = (7, 7)$. In the following exploration experiments the same environment map is built as shown in Fig. 4(a). It has shown that the robot can map the environment with VCF based navigation algorithm. Only the robot tracks are depicted, so that we can concentrate on the robot motion mode as it follows the VCF based navigation algorithm. In fact, the robot motion tracks give a description of the environment information. Since the robot cannot pass through the obstacles, the area surrounded by the robot, but not passed by it, indicates that it is occupied by an obstacle. The fact that the robot's initial position can be placed randomly, but not at the central section, shows that the proposed algorithm has a better universality.

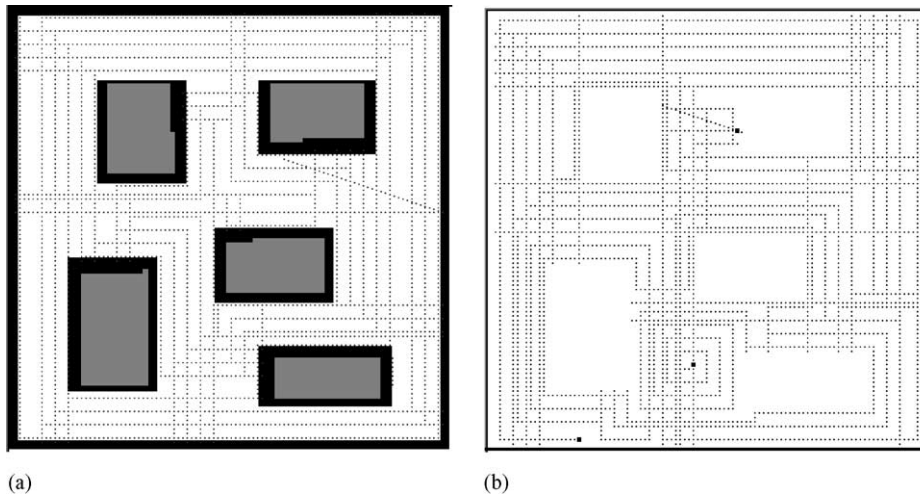


Fig. 4. Robot tracks for environment exploration with different initial positions. (a) Initial position at $r_0 = (7, 7)$. (b) Initial position at $r_0 = (140, 240)$.

In the case where the robot reaches a boundary place, it may explore only a part of the whole environment.

According to the VCF based algorithm, the robot tends to continue exploration towards the undetected area. If the robot is inside the interior of the explored section, it will search for a new position namely the frontier which is between the explored and unexplored areas and continue another round of exploration from that new initial position till the whole environment is explored completely. In Fig. 4(b) depicts the robot track after fulfilling the exploration task with a random initial position $r_0 = (140, 240)$. A tendency from the detected section toward the undetected section and the obstacle avoidance effect can be seen from the track. The experiments using the VCF based navigation algorithm in this section show that for different initial positions the search modes and tracks may be different, but the same results can be obtained, that is, mapping the environment entirely.

The initial position is not important for the robot exploration. It can even take its initial position as the origin of the reference frame and compute its position at any time by way of the odometry method.

In the exploration tasks, the environments are usually closed with walls. When a robot detects those closed walls, which construct its exploration region boundaries, it knows, that is its exploration boundaries. With the variable $rec(i, j)$ distinguishing the detected

cells from the undetected ones, the robot can judge if all the reachable areas have been explored.

3.3. Simulation experiment 3: multi-robot exploration with random initial positions

Many works were contributed to single robotic mapping. For large scale and complex environments, cooperative exploration by multiple robots is needed. Up to now, little research work has been done in multi-robot mapping. The territorial section segmentation strategies are usually adopted, that is, each robot takes charge of some sections to build local maps and then all the local maps are combined into a whole one. Such task distribution has little difference from the single robot mapping in essence. Very little cooperation between robots is needed as little interference happens during the segmentation exploration. A good exploration strategy by multiple robots is that the exploration tasks are distributed appropriately along with the exploration process by the way of the on-line cooperation between robots so that the environment can be mapped with good accuracy in a short time. Good cooperative strategies can bring out the advantages of multi-robot systems. In this research, the robots using the VCF based navigation algorithm may be close to each other in their exploration and their detecting areas often interleave. Generally, for the case where the detecting areas of different robots often interleave or adjoin,

collision avoidance between the robots and repetitive detection avoidance are significant issues in addition to collision avoidance between the robot and the obstacle. With information share among the robots, the rules of the VCF based navigation algorithm will give a solution to the problems. Exploration distribution among the robots can be implemented cooperatively on-line and no special cooperation strategies needs to be designed.

On the basis of exploration by a single robot with a random initial position, the experiment on the multiple robot mapping is carried out using the VCF based navigation algorithm. Two robots are employed to carry out the mapping task. The robots can communicate with each other and they can share all the mapping information. This means that the environment information obtained in the sensor range of one robot is also known to the other robot and that region is recorded as “detected” by both robots. Following the VCF based navigation algorithm, the other robot will not walk too close to the robot and collisions between the robots are avoided. Still take Fig. 3(a) as the experiment environment, the initial positions of the two robots are set as $r_{10} = (140, 160)$ and $r_{20} = (140, 240)$, respectively. When the exploration is finished, the tracks passed by the two robots are recorded together as shown in Fig. 5(a). The initial and termination positions of both robots are also marked as S_{r1} , T_{r1} , S_{r2} , T_{r2} , respectively in Fig. 5(a). The tracks generated by robot r2 are shown

in Fig. 5(b). It can be seen from Fig. 4 that the tracks of both robots are completed, which means that the robots share the exploration information and thus improve the operation efficiency. It also shows that the distributed multi-robot structure adopting the VCF based navigation algorithm is advantageous to the multi-robot system that uses the territorial segmentation strategy.

In respect of the exploration steps, the step sum of multiple robots has no advantage over a single robot. But the parallel detection process decreases the total exploration time to increase the system efficiency.

3.4. Simulation experiment 4: exploration of the environment where concave obstacle exists

In the above simulation experiments, all the obstacles in the environment are convex. In fact, the VCF navigation algorithms can also allow robots to explore environments where concave obstacles exist. To demonstrate that, the following simulations are executed where concave obstacles exist in the environments. Fig. 6(a) shows the robot exploration tracks in the environment where concave obstacles exist. In Fig. 6(b) is the environment map generated after exploration. In case of the partially explored regions, where there might be multiple boundary regions, the VCF navigation algorithm makes the robot exploration to have the same effect.

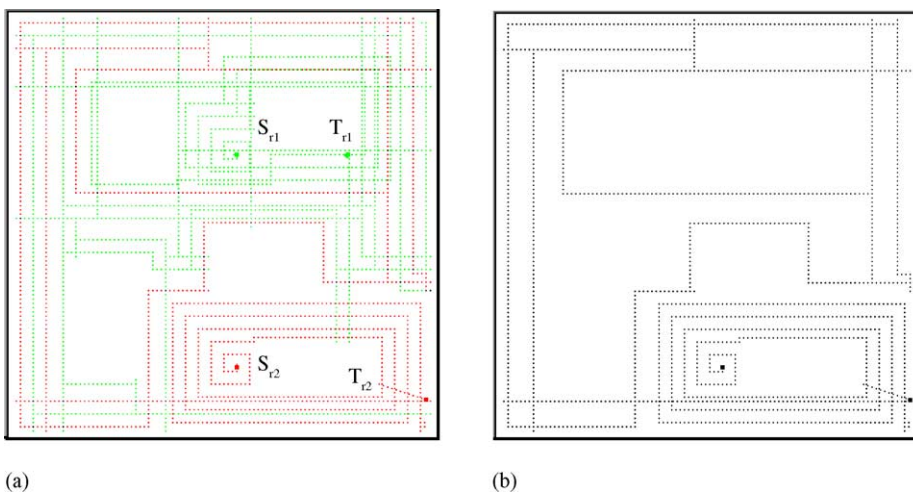


Fig. 5. Tracks of multi-robot exploration. (a) Tracks generated by both robots r1 and r2. (b) Tracks generated by r2.

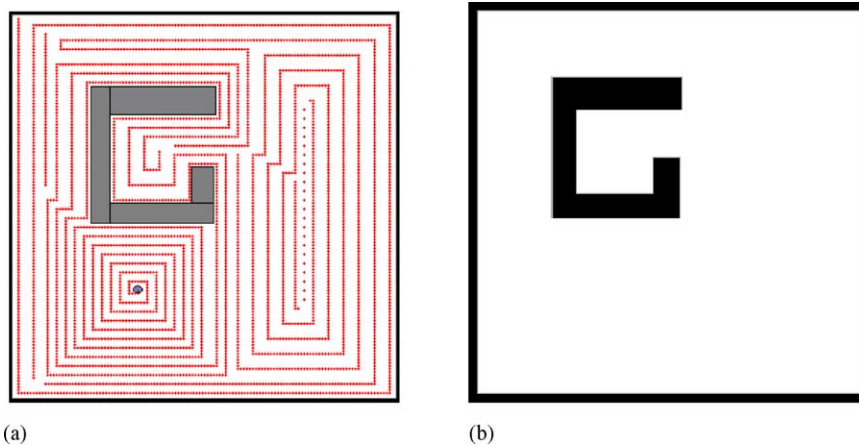


Fig. 6. Exploration of the environment where concave obstacle exists. (a) Exploration racks of the robot in concave environment. (b) Map generated by the robot.

3.5. Simulation experiment 5: exploration and trash collection by multiple robots

Trash collection is usually a concurrent task for robot in the environment exploration. Here experiments will be simulated to complete the concurrent task of exploration and trash collection by multiple robots adopting the VCF algorithm. Several trashes are scattered in the environment given in Fig. 3(a), as shown in Fig. 7(a). In this experiment, two robots are employed to map the environment depicted in Fig. 7(a) and to clean up the trashes in the environment. Robot r1 takes charge of constructing map for the environment and r2 takes charge of collecting the trashes. For mapping, robot r1 uses the VCF based navigation algorithm.

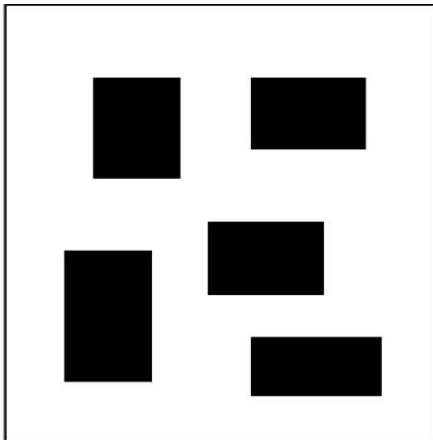
During this experiment, the two robots have to solve the following two issues: to avoid collisions between the robots and obstacles and collisions between the robots; to plan reasonable paths for robot r2 to collect the trashes. As for the question of avoiding collisions between robot r1 and the environment obstacles, the solution has been contained in the navigation algorithm per se. For the collision avoidance between robots r1 and r2, assume that robot r1 has priority over robot r2 in

their motion. This means if the two robots detect that collision will happen, the strategy to be taken is that robot r1 continues its motion while robot r2 waits until robot r1 passes.

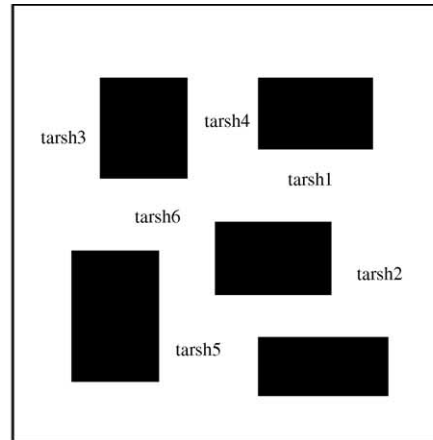
In this experiment both robots r1 and r2 are placed near the central section of the environment. Robot distinguishes trashes by their occupied area. Usually the trash is discrete and occupies no more than a whole cell. Robot r1 marks the trashes with numbers in their detecting order. Robot r2 remains at its initial position until robot r1 informs that the first trash is detected. In this way, robot r2 cleans the trashes one by one in the order that the trashes are numbered. For the trash collection, robot r2 is required to do the following: (1) reaching the target trash as quickly as possible at a constant speed, that means it should pass a distance as short as possible; (2) avoiding collision with obstacles; (3) avoiding collision with robot r1. For the former two requirements, a path-planning algorithm is designed, based on the obstacle influence factor. The algorithm is described as follows.

At each current position, robot r2 has four possible directions to select for its next step: turn right at 90° , turn right at 180° , turn left at 90° , and keep the current

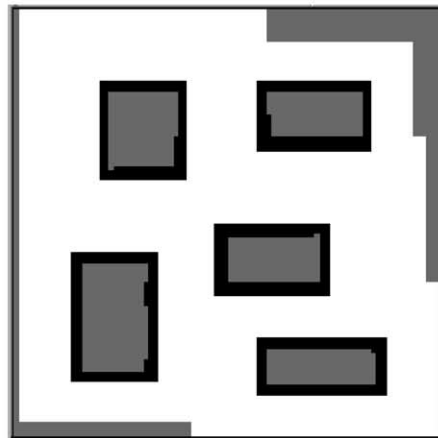
Fig. 7. Trash collection by multiple robot in unknown environment. (a) Environment with trashes to be collected. (b) Trash marked in the order that they are detected by r1. (c) Shows the environment when r1 finish its exploration; some of the trash has been cleared by r2. (d) Shows the trash-collecting track of r2 when r1 finishes its exploration. (e) Shows the trash-collecting track of r2 when it has collected all the trashes detected by r1 one by one. (f) Shows the mapped environment with trashes cleared completely, the terminating positions of r1 and r2 are also marked.



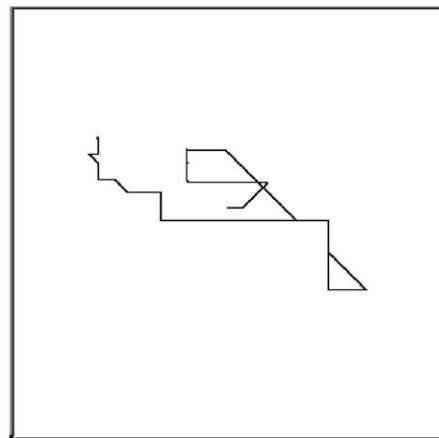
(a)



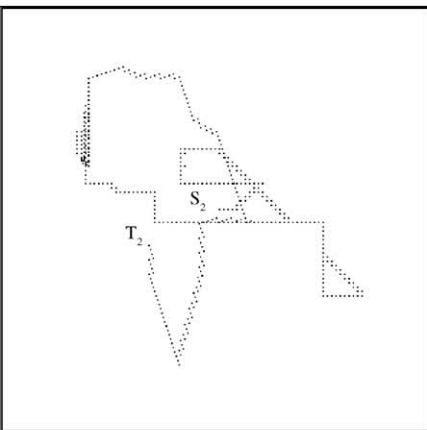
(b)



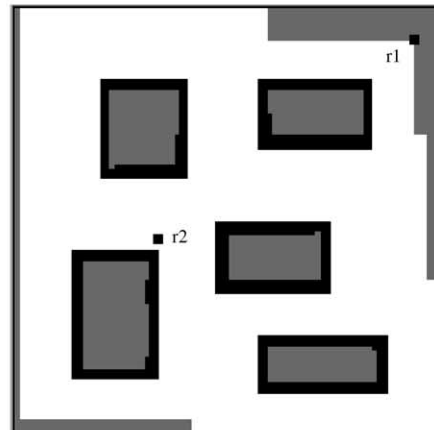
(c)



(d)



(e)



(f)

direction unchanged. The corresponding positions for the next step are: $(r2.x + \text{step2} \times \cos(\alpha + 90^\circ), r2.y + \text{step2} \times \sin(\alpha + 90^\circ))$, $(r2.x + \text{step2} \times \cos(\alpha + 180^\circ), r2.y + \text{step2} \times \sin(\alpha + 180^\circ))$, $(r2.x + \text{step2} \times \cos(\alpha - 90^\circ), r2.y + \text{step2} \times \sin(\alpha - 90^\circ))$, and $(r2.x + \text{step2} \times \cos(\alpha), r2.y + \text{step2} \times \sin(\alpha))$, where step2 represents the step length of robot r2, and α represents the direction of robot r2 at its current step, equalling the angle measured from X -axis to the moving direction of robot r2. Define an evaluation function V :

$$V = \kappa \cdot D^2 + \lambda \cdot \omega, \quad (7)$$

where κ, λ are the adjustable weights whose different values will influence the path generated by robot r2 and D represents the distance from a possible position $\text{cell}(x, y)$ to the target trash.

$$D = \sqrt{(\text{cell}.x - \text{target}.x)^2 + (\text{cell}.y - \text{target}.y)^2}. \quad (8)$$

where ω represents the influence factor of $\text{cell}(x, y)$ to the decision of the robot r2. A good next position should have a low V value. In this research it denotes the state of a candidate cell. If the $\text{cell}(x, y)$ is occupied by an obstacle, the value of ω is a positive infinity, and if the cell is free, the value of ω is 0, that is,

$$\omega(\text{cell}) = \begin{cases} 0 & \text{free} \\ +\infty & \text{occupied} \end{cases}. \quad (9)$$

The third request can be satisfied by setting motion priorities. When robot r2 finds that at the selected next position it will collide with robot r1, robot r2 stops its motion, i.e. $\text{step2} = 0$. Robot r2 will wait at its current position while robot r1 continues its exploration. Robot r2 restores its motion after robot r1 passes by.

The black blocks in Fig. 7(a) represent the environment obstacles, and the black points represent the trashes scattered in the environment. The trashes and their positions that they are detected by robot r1 are recorded as follows: $\text{trashrec}[1]-(180, 120)$, $\text{trashrec}[2]-(250, 200)$, $\text{trashrec}[3]-(50, 100)$, $\text{trashrec}[4]-(150, 90)$, $\text{trashrec}[5]-(120, 250)$, $\text{trashrec}[6]-(100, 160)$. In Fig. 7(b) these environment trashes are marked in the order they are detected by robot r1. Fig. 7(c) shows the environment map built by robot r1 when it completes the exploration, and Fig. 7(d) shows the track of robot r2 collecting trashes

one by one using the information recorded by robot r1. It can be seen that when robot r1 completes its exploration, robot r2 has cleaned some of the trashes. Fig. 7(e) demonstrates the path track of robot r2 after it collects all the trashes in the environment. Fig. 7(f) illustrates the environment features and the termination positions of robots r1 and r2.

4. Conclusions and discussions

In this paper a real-time navigation algorithm based on virtual centrifugal force has been proposed and applied to the explorative task simulations for validation. Robots can build an integrated map of the environments by navigating in a mode similar to Archimedes spiral so that the main features of the environment can be detected. The kinematics and dynamics equations of the robot adopting VCF algorithm have been generated. Moreover, using this algorithm collision with obstacles can be avoided so as to lay a foundation for more tasks in unknown environment.

Several simulation experiments have been carried out to verify the algorithm on single and multiple robots for environment explorations and multiple robot trash collection. Compared with the previous navigation algorithms, this algorithm has the function of avoiding collisions and the capability of navigation, in addition to meeting the requirements of exploring unknown terrains. There is no special need about robots or environment conditions to use this algorithm and the algorithm has good applicability. Its succinctness means shorter time and less memory space occupancies for robot controllers to make decisions, and consequently improves the efficiency of robotic task implementations. The simple data structure makes the computation time short enough for the robots to explore the environment on-line in real-time.

In the experiments with real robots, it is quite possible that there exist sensing and localization uncertainties. Sensors and techniques in mobile robot positioning are reviewed in [26]. Most sensing uncertainties can be decreased in virtue of data fusion theories after modelling the sensor uncertainties [14,15]. Also the localization uncertainties can be decreased by dint of filter methods so that the map can be built correctly. Localization algorithms such as Kalman filters [27–29], particle filters [13,30,31] and so on have been

successfully applied in a range of real robot systems. Two well-known localization platforms are Minerva and Rhino. The localization errors will not affect the complete exploration of the whole environment except that the representation map will be built with distortion.

To extract topological map when the robots navigate in the unknown environment is our future research goal.

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